# CS 228: Biometric Security with AI

Fall 2026, Dr. Amith Kamath Belman

## Project Checkpoint Assignment 1

**Due: Thursday, Oct 27, 2026**

**One submission per team (or per individual for a one person project). No late submissions.**

**Total 100 points;**

**Expected submission:** a 3 to 4 page technical report as a PDF (IEEE or ACM LaTeX template) and a link to a working code repository.

**AI usage:** You may use an AI assistant for coding and for writing the document, but do not paste in anything you have not read and approved. Everything you submit is your work, and you will be asked to explain and elaborate on it. A part of the work that you cannot explain when questioned is treated as plagiarism.

### 1. Overview

The checkpoint is the first technical deliverable in the project. By the due date you must have secured your dataset, explored it, and measured a simple baseline. Each team submits one PDF report and a link to a functional code repository, and the report must follow the format below.

### 2. The technical progress report

Write the report in any one of these four templates (3 to 4 pages including figures and tables; references do not count toward the length):

- ACM single column: https://www.overleaf.com/latex/templates/acm-conference-proceedings-primary-article-template/wbvnghjbzwpc
- ACM double column: https://www.overleaf.com/latex/templates/association-for-computing-machinery-acm-large-2-column-format-template/qwcgpbmkkvpq
- IEEE single column: https://www.overleaf.com/latex/templates/one-column-ieee-journal-article/ykctxqgxptrs
- IEEE double column: https://www.overleaf.com/latex/templates/ieee-conference-template/grfzhhncsfqn

The cover block (title, team members, and the nature of the project in one line: build, attack, or test) does not count toward the length. The report must contain the following four sections; each is worth 20 points, and the repository in Section 3 is the fifth 20.

#### I. Problem statement and motivation (20 points)

- Give a precise definition of the problem you are solving.
- Explain why the problem matters in the current security landscape (for example spoofing, data poisoning, adversarial examples, continuous authentication, or demographic differentials in error rates).
- If your project scope has shifted since the proposal report, or since the questions at your proposal presentation, highlight the changes in a different text color.
- State the nature of the project in one line, as in the proposal: build robust X, attack X, or test the feasibility of Y.

**II. Data engineering (20 points)**

Describe:

- **Dataset origin:** where the data came from (a public repository with its license and access terms, web collection, or your own collection with the consent of the people recorded), with a citation.
- **Dataset characteristics:** number of subjects, samples per subject, sessions, image resolution or signal sampling rate, and the labels (identity, genuine or impostor, live or spoof). Adjust the list to your data.
- **Cleaning and preprocessing:** the steps taken for missing values, normalization, alignment or cropping, and feature reduction (PCA, correlation based selection, and so on), with the reason for each step.
- **Schema sanity and quality:** outliers, anomalies, duplicates, or mislabeled samples found at the first load, and the shortcomings of the data for your problem.
- **The split :**state the split and data used clearly.

**III. Exploratory data analysis (20 points)**

Provide some suitable visualization for the data:

- **Univariate analysis:** histograms and box plots of key features (or of image statistics, signal lengths, and quality scores when the raw data are images or signals).
- **Bivariate analysis:** a correlation matrix or scatter plots showing how features relate to each other and to the labels.
- **Class balance:** a visual of the class distribution (subjects, genuine against impostor, live against spoof) and what it implies: oversampling, balanced impostor sampling, or per class metrics.
- Every figure must a title, labeled axes, and one sentence on what you want to express through it.

**IV. Baseline implementation and preliminary results (20 points)**

Implement a simple baseline (majority class, k-NN, logistic regression, a small pretrained CNN, or the reference method your three papers compare against) and report:

- **Performance metrics, with the count for every rate.**
- **Threshold analysis:** where a threshold exists, the EER or a plot of FAR against FRR, and the operating point you would choose and why.
- **Error analysis:** which kinds of samples the baseline fails on, with a hypothesis for why.
- Fix and report your random seeds; every number in this section must be reproducible from the repository at the submitted commit.

**3. Code repository submission (20 points)**

Include a link to a GitHub or GitLab repository that can be used by me:

- a README.md with instructions for setting up the environment.
- documented Python scripts or Jupyter notebooks for data loading, the EDA, and the baseline training and evaluation;
- a list of dependencies (requirements.txt or an environment file);
- no dataset in the repository: commit the script that fetches it, or the instructions for placing it.